Pacman Game Project Report

Executive Summary

The Pacman Game project is a Java-based implementation of the classic arcade game, developed using object-oriented programming (OOP) concepts in IntelliJ IDEA. This report outlines the project's objectives, design, implementation, and future improvements.

Introduction:

The Pacman game demonstrate the concept of OOP like Association, classes, packages and provide encapsulation through data hiding and abstraction.

Objectives:

1. Implement the concept of classes.
2. Implement relationship between them.
3. Design simple game with the help of pacakges.
4. Control the movement of pacman and ghost.

Tools and Technologies:

- **Programming Language:** Java
- **IDE**: IntelliJ IDEA
- **Version Control:** Git & GitHub

Project Structure:

The project is divided into packages:

1. **main package**
    a. It contains the main game execution class: main.java.
2. **person package**
    a. It contains the class: pacman.java.
3. **ghost package**
    a. It contains ghost class: ghost.java.
4. **map package**
    a. Contain the gamemap class which is used to implement main logic.

System Design:

Main Components

1. **Game Class**

- Control the main class
- Start and manage game
- Create association between different classes.
- Implement logic

2. Pacman Class

- Represent pacman.
- Handles direction of pacman.
- It has a strong composition with gamemap class.
- Display and calculate score.

3. Ghost Class

- Represent ghosts/enemies.
- Update ghost position randomly.
- It also has a strong association with gamemap class.

4. Gamemap Class

- It control game grid, food ,and walls of the game.
- Check the validity of the user input.

Working of the System:

1. The game starts from the `Game` class
2. A map is created using `Gamemap`
3. Pacman is initialized at a starting position
4. Ghosts are created and placed on the map
5. Game loop runs:
    a) Pacman moves based on input
    b) Ghosts move randomly
    c) Collisions are checked
6. Game continues until a stopping condition

Key Features:

1. Modular design using packages
2. Random ghost movement using `Random` class
3. Clear separation of responsibilities
4. Simple and readable code structure

Concepts Used:

1. Classes and Objects
2. Encapsulation
3. Packages
4. Random number generation
5. Basic game loop logic

Challenges Faced:

1. Managing multiple classes and packages
2. Handling movement logic
3. Ensuring correct interaction between objects

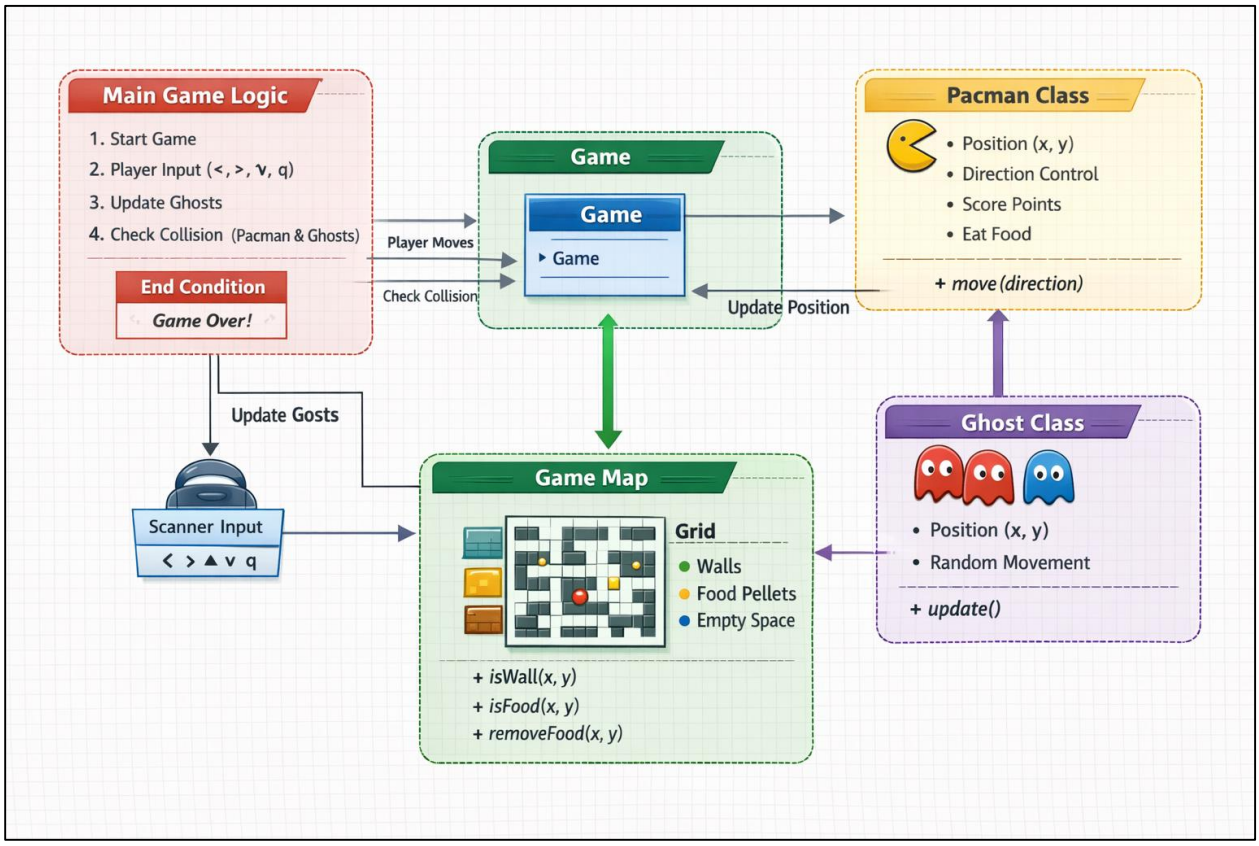
Future Improvements:

1. Add GUI using Java Swing or JavaFX
2. Implement scoring system
3. Add smarter ghost AI
4. Add levels and difficulty increase

Conclusion:

This project successfully demonstrates OOP concepts in building a simple game, providing a foundation for more advanced game development and improving understanding of Java programming and project structuring.

References

- Java Documentation
- IntelliJ IDEA Documentation
- Online programming resources

Architectural Diagram